Team: TEAM NUMBER REQUIRED

Problem B

Predicting a Penalty Kick from 0.10 Seconds of Motion

Abstract

Two early positions of a penalty kick are given, but the wind, air density and drag coefficient are not measured. This paper asks what the rest of the flight can look like under one clear force model. We use gravity, quadratic drag relative to the air, and an effective sideways force written as a cross product, all acting from the kick to the goal plane. The sideways term is not a measurement of spin; it combines spin and lift. For each listed wind, density and drag value we fit the launch velocity and two sideways coefficients to the two positions. A fit is accepted only when all six coordinate residuals lie within a stated allowed error. The lists contain 210 parameter choices; 86 are accepted at 5 mm. Their goal-plane coordinates span $y \in [2.5384, 2.6812]$ m and $z \in [1.9451, 1.9867]$ m, and two accepted trajectories differ by 0.1477 m. For the current example at wind 8.0 m s^{-1}, density 1.225 kg m^{-3} and $C_D = 0.350$, the centre crosses the plane at $(2.6087, 1.9598)$ m after 0.4042 s. The spread shows that the two observations do not select one landing point; it is not a probability. MATLAB checks use three step sizes, a work-energy balance and event tests. No independent measured full flight is available, so experimental confirmation remains to be done. Only the listed parameter values are searched, and the ranges in the figures do not guarantee that all possible cases are included.

1 Problem and approach

This section states what we compute and in what order to read the paper.

The ball centre is given at the kick, $\mathbf{r}_0 = (0, 0, 0.11)$ m, and at $t_1 = 0.05$ s and $t_2 = 0.10$ s:

$$\mathbf{r}_1 = (1.50, 0.45, 0.44) \text{ m}, \qquad \mathbf{r}_2 = (2.96, 0.86, 0.74) \text{ m}. \tag{1}$$

We take x toward the goal, y across the goal and z upward. These six numbers are the only position data. Wind, density, drag and launch conditions are unknown, so one trajectory is not determined. We ask instead: for a listed set of environmental values, which launch conditions reproduce both positions within an allowed error, and where do those trajectories cross $x = 11$ m? The reading order is as follows: Section 2 writes the force law, Section 3 explains how the observations select parameter choices, Section 4 gives the numbers and figures, Section 5 lists checks and limits, and Section 6 answers the problem.

A trajectory is retained when it meets the allowed coordinate error and the physical checks. Figure 1 shows the calculation: choose an environment, solve for launch conditions, and check the resulting flight. Rejected cases are recorded alongside accepted ones.

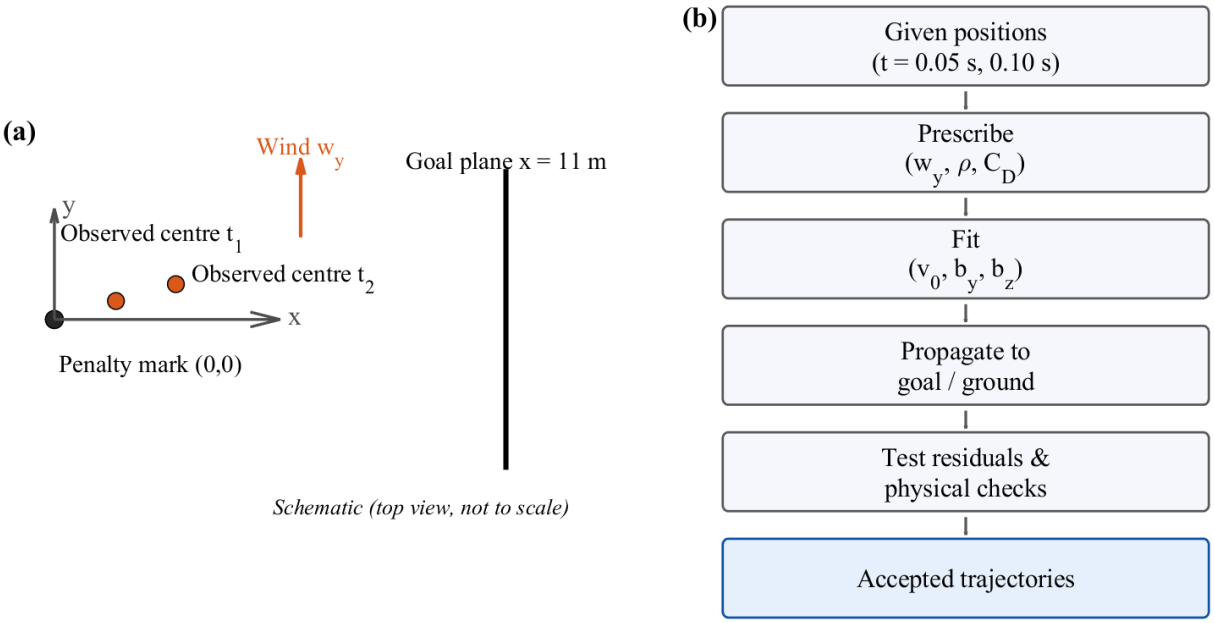


Figure 1: Coordinate convention and the path from two observations to accepted trajectories. The left panel shows the horizontal geometry; the right panel shows listed environment values, fitted launch conditions, the allowed-error and physical checks, and the recorded terminal events.

2 Physical model from kick to goal

This section builds the force law in order: define the moving system, relate airflow to force, combine the forces, and check the size and limits of the approximations.

2.1 System, coordinates and initial conditions

The system is the ball after release from the foot and before contact with the ground or a post. Its centre is $\mathbf{r} = (x, y, z)$ and $\mathbf{v} = \dot{\mathbf{r}}$; x points toward the goal and z upward. With radius $R = 0.11$ m and mass $m = 0.43$ kg,

$$\mathbf{r}(0) = (0, 0, R), \quad \mathbf{v}(0) = \mathbf{v}_0, \quad \mathbf{g} = (0, 0, -9.81) \text{ m s}^{-2}. \tag{2}$$

The chosen mass and circumference $2\pi R$ satisfy IFAB specifications [2]; they are not measurements of this particular ball. The kick is represented by the unknown launch velocity. After release there is no continuing force from the foot. Newton's second law for the centre is

$$m\dot{\mathbf{v}} = \mathbf{F}_g + \mathbf{F}_D + \mathbf{F}_M, \qquad \mathbf{F}_g = m\mathbf{g}. \tag{3}$$

The next steps establish drag $\mathbf{F}_D$ and the transverse force $\mathbf{F}_M$ instead of prescribing a fitted acceleration curve.

2.2 Drag from relative airflow

For steady lateral wind $\mathbf{w} = (0, w_y, 0)$ define

$$\mathbf{u} = \mathbf{v} - \mathbf{w}, \quad U = \|\mathbf{u}\|, \quad A = \pi R^2, \quad p_{\text{dyn}} = \tfrac{1}{2}\rho U^2. \tag{4}$$

Dynamic pressure multiplied by frontal area gives a force scale. The dimensionless drag coefficient includes effects of the surface, boundary layer and flow separation [4]. Hence

$$D = p_{\text{dyn}} A C_D, \qquad \mathbf{F}_D = -D\frac{\mathbf{u}}{U} = -\tfrac{1}{2}\rho A C_D U \mathbf{u}. \tag{5}$$

The sign makes drag oppose relative motion, and $\rho A U^2$ has units of newtons. Wind changes this same force through $\mathbf{u}$; it is not an additional copy of drag. Dynamic pressure supplies a scale, while C_D requires calibration. Bernoulli's equation or conservation of energy alone cannot determine its numerical value.

2.3 Meaning of the effective transverse force

Rotation changes the boundary layer and wake and can produce a force perpendicular to airflow. Football wind-tunnel measurements support this mechanism while showing dependence on rotation and flow regime [3]. Write its magnitude as

$$F_M = \tfrac{1}{2}\rho A U^2 C_L. \tag{6}$$

Let $\boldsymbol{\omega}_\perp$ denote the angular-velocity component perpendicular to $\mathbf{u}$. Over a limited range, assume $C_L \simeq c_s R\|\boldsymbol{\omega}_\perp\|/U$ with dimensionless slope c_s and direction along $\boldsymbol{\omega} \times \mathbf{u}$. Substitution gives

$$\mathbf{F}_M \simeq \tfrac{1}{2}\rho A c_s R(\boldsymbol{\omega} \times \mathbf{u}) = m(\mathbf{b} \times \mathbf{u}), \qquad \mathbf{b} = \frac{\rho A c_s R}{2m}\boldsymbol{\omega}. \tag{7}$$

The linear coefficient relation is a modelling approximation, not a universal law established by the cited experiment. The fitted $\mathbf{b}$, in s^{-1}, combines spin with unknown aerodynamic efficiency. Two positions cannot identify these separately, so $\mathbf{b}$ is not reported as a measured angular velocity.

We set $\mathbf{b} = (0, b_y, b_z)$: for predominantly forward motion, b_z controls lateral bending and b_y vertical bending. Fixing $b_x = 0$ reduces the unknowns but does not prove that the actual component vanishes. Resolving spin would also require inertia and aerodynamic torque in $\mathbf{I}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}$, which are not given. Constant $\mathbf{b}$ is a short-flight approximation whose error remains part of the model limitations.

2.4 The complete equations used for integration

Substituting the two force expressions into Newton's law and defining $K = \rho A/(2m)$ yields six first-order equations:

$$\begin{aligned} \dot{\mathbf{r}} &= \mathbf{v}, \qquad \dot{\mathbf{v}} = \mathbf{g} - KC_D U\mathbf{u} + \mathbf{b} \times \mathbf{u}, \\ \dot{v}_x &= -KC_D U v_x + b_y v_z - b_z(v_y - w_y), \\ \dot{v}_y &= -KC_D U(v_y - w_y) + b_z v_x, \\ \dot{v}_z &= -g - KC_D U v_z - b_y v_x, \qquad g = \|\mathbf{g}\|. \end{aligned} \tag{8}$$

These components explain coupling: wind changes both lateral drag and the x component of the transverse force. Changing one input therefore requires recomputing the complete state.

Table 1: Symbols used in the force derivation.

Symbol	Meaning	Unit or role
$\mathbf{r}, \mathbf{v}, \mathbf{u}$	Centre, ground velocity, relative velocity	m; m s^{-1}; m s^{-1}
m, R, A	Mass, radius, frontal area	kg; m; m^2
ρ, w_y, C_D	Density, wind, drag coefficient	prescribed scenarios
$\mathbf{v}_0, b_y, b_z$	Launch and effective transverse coefficients	fitted; m s^{-1}; s^{-1}
$K, C_L, C_\perp$	Aerodynamic scale and force coefficients	m^{-1}; dimensionless

2.5 Scale checks and approximation limits

For the current example, launch relative speed is $U_0 = 31.26$ m s^{-1} and flight time $T = 0.4042$ s. The initial weight, drag and transverse-force magnitudes are approximately 4.22, 7.96 and 6.96 N. Aerodynamic forces are comparable to weight. Their relative velocity changes over one flight are estimated by

$$\Pi_D = KC_D U_0 T = 0.239, \quad \Pi_M = \|\mathbf{b}\|T = 0.215, \quad \Pi_g = \frac{gT}{U_0} = 0.127. \tag{9}$$

None is very small, so all three accelerations are retained. With an assumed viscosity $\mu = 1.81 \times 10^{-5}$ Pa s and sound speed $c = 343$ m s^{-1},

$$\mathrm{Re} = \frac{2\rho R U_0}{\mu} \simeq 4.65 \times 10^5, \qquad \mathrm{Ma} = \frac{U_0}{c} \simeq 0.091. \tag{10}$$

Viscosity and sound speed are scale assumptions, not site measurements. The small Mach number supports neglecting compressibility. Reynolds number makes the dependence of C_D on flow regime relevant; it does not calibrate a constant C_D for this ball.

Neglected terms also deserve scale estimates. The buoyancy-to-weight ratio is $\rho(4\pi R^3/3)/m \simeq 1.59\%$. Without refitting launch conditions, omitting buoyancy gives a vertical-displacement scale of about 1.27 cm over this flight. Thus millimetre fitting residuals cannot be called millimetre physical accuracy. With Earth's rotation rate $\Omega = 7.292115 \times 10^{-5}$ s^{-1}, the displacement scale $\Omega\|\mathbf{v}_0\|T^2$ of roughly 0.39 mm, below the chosen coordinate tolerance. Buoyancy, changing wind and changing coefficients remain sources of model error; additional observations are needed to test them.

2.6 Work and energy in moving air

For $E = \frac{1}{2}m\|\mathbf{v}\|^2 + mgz$, taking the scalar product of Newton's law with ground-frame velocity cancels the gravitational potential term:

$$\begin{aligned}\frac{dE}{dt} &= (\mathbf{F}_D + \mathbf{F}_M)\cdot\mathbf{v} \\ &= -\tfrac{1}{2}\rho A C_D U^3 + (\mathbf{F}_D + \mathbf{F}_M)\cdot\mathbf{w}.\end{aligned} \tag{11}$$

Here $\mathbf{F}_M \cdot \mathbf{u} = 0$, but $\mathbf{v} = \mathbf{u} + \mathbf{w}$. The transverse force is perpendicular to relative airflow, yet it can do ground-frame work when wind is present. Ground-frame mechanical energy therefore need not decrease monotonically. Integrating Eq. (11) supplies the work-energy check used later.

3 Finding parameters from the observations

This section explains how the two positions become parameter choices.

For each listed triple $\eta = (w_y, \rho, C_D)$ we fit $q = (v_{0x}, v_{0y}, v_{0z}, b_y, b_z)$ by bounded nonlinear least squares,

$$q^*(\eta) = \arg\min_{q\in\mathcal{B}} \sum_{i=1}^{2} \|\mathbf{r}(t_i; q, \eta) - \mathbf{r}_i\|^2, \tag{12}$$

using `lsqnonlin` [6] from two starting points. The fit uses fourth-order Runge–Kutta integration with step 0.0025 s. Acceptance is re-evaluated with `ode45` at RelTol=10^{-11} and AbsTol=10^{-12}; whole-flight propagation uses RelTol=10^{-9}, AbsTol=10^{-11} and maximum step 0.002 s. A fit is checked on all six coordinates,

$$|r_j(t_i; q, \eta) - r_{ij}| \le \epsilon, \qquad i = 1, 2, \quad j = x, y, z, \tag{13}$$

at $\epsilon = 2, 5, 10, 20$ mm; 5 mm is the default. The allowed error is a stated scenario, not a measured camera accuracy.

The search uses 7 wind values on $[-8.000, 8.000]$ m s^{-1}, 3 density values $\{1.100, 1.225, 1.350\}$ kg m^{-3}, and 10 drag values on $[0.15, 0.60]$, giving 210 parameter choices. These values are example settings, not measured distributions. Only the listed parameter values are searched, and the ranges in the figures do not guarantee that all possible cases are included.

Before a fit is called accepted it also passes a physical screening. The launch must point forward and not downward, the initial speed must lie between 10 and 40 m s^{-1}, and along the sampled trajectory the side-force coefficient must satisfy $C_\perp \le 0.50$. The ball must stay above the ground before the terminal event, and the event must be located within 2 s. An accepted fit must pass both the six-coordinate test and this screening; observation fit and physical screening are stored separately. The speed range and the 0.50 cap are conservative model screens, not universal limits for players or footballs. We do not force the crossing into the goal: a miss is a recorded outcome, and the goal is used only to describe where the ball goes.

Two further points explain why the result is a set rather than one shot. First, the fitting problem has more unknowns than the six coordinates. Changing the wind or the drag changes the air force, and a different launch velocity can make up for much of that change at the two observed times. The fit therefore finds different launch conditions for different listed environments, even though all of them look the same at $t = 0.05$ and $t = 0.10$ s. Second, the two observations constrain the early path only, and a small difference in the fitted sideways coefficient keeps acting after the last observation. The accepted set is the result of searching the listed values, not a proof that nothing outside the grid exists.

3.1 How different parameters can meet the same observations

The screening has a mathematical explanation. Collect the unknowns and observation map as

$$p = (v_{0x}, v_{0y}, v_{0z}, b_y, b_z, w_y, \rho, C_D), \qquad \mathbf{f}(p) = \begin{bmatrix} \mathbf{r}(t_1; p) \\ \mathbf{r}(t_2; p) \end{bmatrix}. \tag{14}$$

There are eight entries in p and six observed coordinates. More directly, drag contains only ρC_D: changing density and drag inversely while keeping wind and effective $\mathbf{b}$ fixed preserves the complete force field. The two values cannot be measured separately from these positions. Changing wind can instead be compensated by different launch conditions within the allowed coordinate error, while later paths differ.

Near a fitted point, let $J = \partial \mathbf{f}/\partial p$. Then

$$\delta \mathbf{f} \simeq J \delta p, \qquad \|\mathbf{f}(p) - \mathbf{r}_{\text{obs}}\|_\infty \leq \epsilon. \tag{15}$$

Changes giving small $J\delta p$ are difficult to distinguish in the early data. This gives a precise local meaning to parameter compensation. Acceptance still requires substitution into the nonlinear equations. Least squares minimizes a sum, not the largest coordinate error, so componentwise screening remains essential. Failure of a fitted choice does not establish that all possible launch conditions in that environment fail.

3.2 Locating the goal point during continuous flight

We do not read the endpoint at an arbitrary common time. The crossing satisfies

$$x(t_g; p) = 11, \quad v_x(t_g) > 0, \qquad (y_g, z_g) = (y(t_g; p), z(t_g; p)). \tag{16}$$

Descending ground contact $z(t) - R = 0,\ v_z(t) < 0$ competes with this event, and the first event is recorded. At the centre plane a simple finite-radius opening check is

$$|y_g| < 3.66 - R, \qquad R \leq z_g < 2.44 - R. \tag{17}$$

The predicted point follows the question's centre-crossing convention $x = 11$. This local clearance check is not a complete post-collision calculation. A ground-first endpoint is never projected onto the goal as a crossing.

3.3 How parameter changes reach the goal point

For small changes that keep relative speed nonzero and preserve a forward goal-first crossing, the crossing time changes along with y, z. Differentiating Eq. (16) gives

$$\frac{\partial t_g}{\partial p} = -\frac{\partial x/\partial p}{v_x(t_g)}, \qquad G = \frac{\partial (y_g, z_g)}{\partial p} = S_{yz} - \frac{1}{v_x(t_g)} \begin{bmatrix} v_y \\ v_z \end{bmatrix} S_x. \tag{18}$$

S_x, S_{yz} are position derivatives at the same fixed time t_g. The second term corrects for the changed arrival time. Omitting it confuses a fixed-time position change with a change in the goal-plane crossing.

To compute these derivatives consistently, write $s = (\mathbf{r}, \mathbf{v})$, $\dot{s} = F(s, p)$ and $S = \partial s/\partial p$. Differentiation yields

$$\dot{S} = F_s S + F_p, \quad F_s = \begin{bmatrix} 0 & I \\ 0 & D \end{bmatrix}, \qquad D = -KC_D \left(UI + \frac{\mathbf{u}\mathbf{u}^T}{U} \right) + [\mathbf{b}]_\times. \tag{19}$$

Here $[\mathbf{b}]_\times \mathbf{u} = \mathbf{b} \times \mathbf{u}$. Initially the position derivatives are zero, the velocity derivatives with respect to launch velocity form the identity, and the other entries are zero. Direct parameter effects include

$$\frac{\partial \dot{\mathbf{v}}}{\partial b_y} = (u_z, 0, -u_x)^T, \quad \frac{\partial \dot{\mathbf{v}}}{\partial b_z} = (-u_y, u_x, 0)^T, \quad \frac{\partial \dot{\mathbf{v}}}{\partial w_y} = -D\mathbf{e}_y. \tag{20}$$

These expressions follow directly from the force directions and the definition of relative velocity. MATLAB integrates the variation equations as a diagnostic and checks Eq. (18) against small parameter perturbations of the original flight equations. The largest derivative discrepancy is 1.38×10^{-10} in the adopted SI parameter units. This tests the implementation without assigning probabilities. The reported trajectories continue to be calculated from the nonlinear flight equations.

4 Results

This section reports the current example, two accepted trajectories with different landings, and the screening figures.

For the current example the listed values are wind 8.0 m s^{-1}, density 1.225 kg m^{-3}, $C_D = 0.350$, allowed error 5 mm, and maximum step 0.002 s. The fit gives

$$(v_{0x}, v_{0y}, v_{0z}, b_y, b_z) = (30.4520, 9.4164, 6.9161, -0.0414, -0.5296), \tag{21}$$

with the first three entries in m s^{-1} and the last two in s^{-1}; the prescribed drag coefficient is $C_D = 0.3500$. The largest coordinate residual is 0.915 mm, and the initial speed is 32.616 m s^{-1}. The ball first crosses $x = 11$ m at

$$(y_g, z_g) = (2.6087, 1.9598) \text{ m}, \qquad t_g = 0.4042 \text{ s}. \tag{22}$$

The recorded terminal status is `plane`, and the finite-radius opening check is yes. These numbers belong to this one parameter choice; they are not a unique answer, and no probability is implied.

Table 2 repeats the fit for every listed parameter choice at four allowed errors. Reading the 5 mm row: 86 of 210 choices are accepted, their crossings span the stated y and z ranges, and the Plane and Opening columns count arrivals and opening clearances. Ground-first outcomes are listed separately; they are not silently extrapolated below the ground.

Table 2: Allowed-error screening of the same parameter choices. Ranges are spreads over the sampled values, not uncertainty bounds.

ϵ (mm)	Accepted	Plane	Opening	Ground	y_g (m)	z_g (m)
2	36/210	36	36	0	[2.5779, 2.6227]	[1.9578, 1.9752]
5	86/210	86	86	0	[2.5384, 2.6812]	[1.9451, 1.9867]
10	168/210	168	168	0	[2.4451, 2.7562]	[1.9183, 2.0053]
20	210/210	210	210	0	[2.2802, 2.8152]	[1.8846, 2.0125]

The two displayed examples are taken from the accepted set at 5 mm; their positions in the MATLAB arrays are 13 and 47. Table 3 lists their values. Both reproduce the two positions within 5 mm, yet their crossings differ by 0.1477 m.

The two rows show the compensation directly. They use different wind and drag values, so the aerodynamic force differs, and the fitted launch speeds are not identical. Each shot also takes its own sideways coefficient to keep the same early positions. These differences also affect the rest of the flight and produce a gap at the goal plane. The gap is modest, and that is expected:

these are steady force values fitted to the same two positions, and a more complex force law would not automatically make the answer more certain.

Table 3: Two accepted trajectories with different goal-plane crossings. Wind acts from the kick in both.

	w_y	ρ	C_D	$\lvert v_0 \rvert$	Error (mm)	y_g	z_g	t_g
A	-8.0	1.225	0.2500	32.323	4.834	2.6812	1.9847	0.3906
B	-5.3	1.225	0.4500	32.849	4.999	2.5389	1.9451	0.4099

Units: wind and speed in m s^{-1}; density in kg m^{-3}; plane coordinates in m; time in s.

Figure 2 explains why different goal points can fit the same observations. In the left panel, all six coordinate errors are within the allowed band for both examples. In the right panel, their late paths separate. Their launch conditions compensate for different aerodynamic forces during the measured interval, but this compensation does not keep the later positions identical.

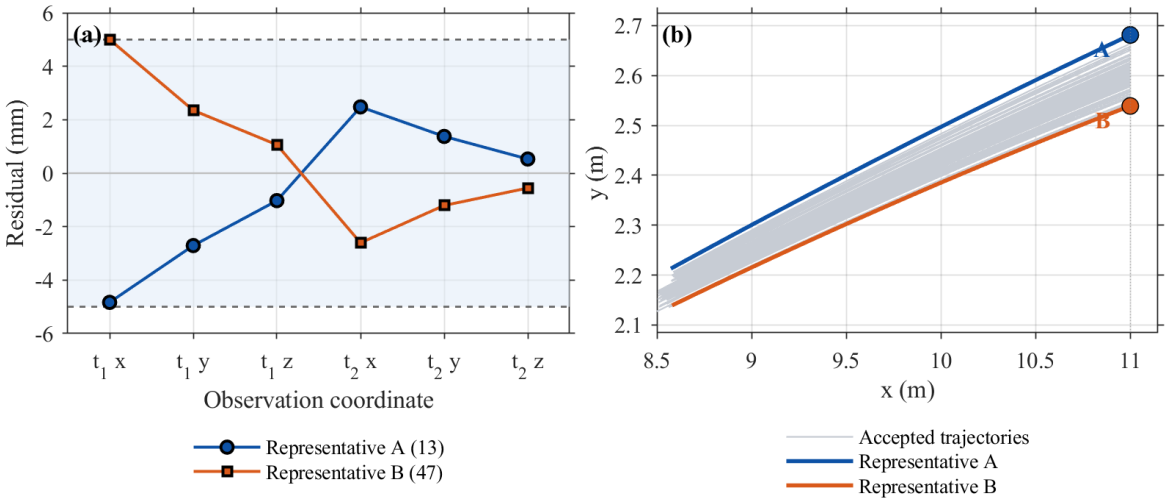


Figure 2: The two accepted trajectories satisfy all six coordinate checks (left), but their late paths separate in the horizontal plane (right). The shaded band is the stated allowed error; displacements are shown at true scale.

Figure 3 shows how the inputs and fitted quantities change together. A larger density–drag product generally requires a larger launch speed for these observations (left). The sideways coefficient changes with wind (middle). Increasing the allowed measurement error accepts more parameter choices (right).

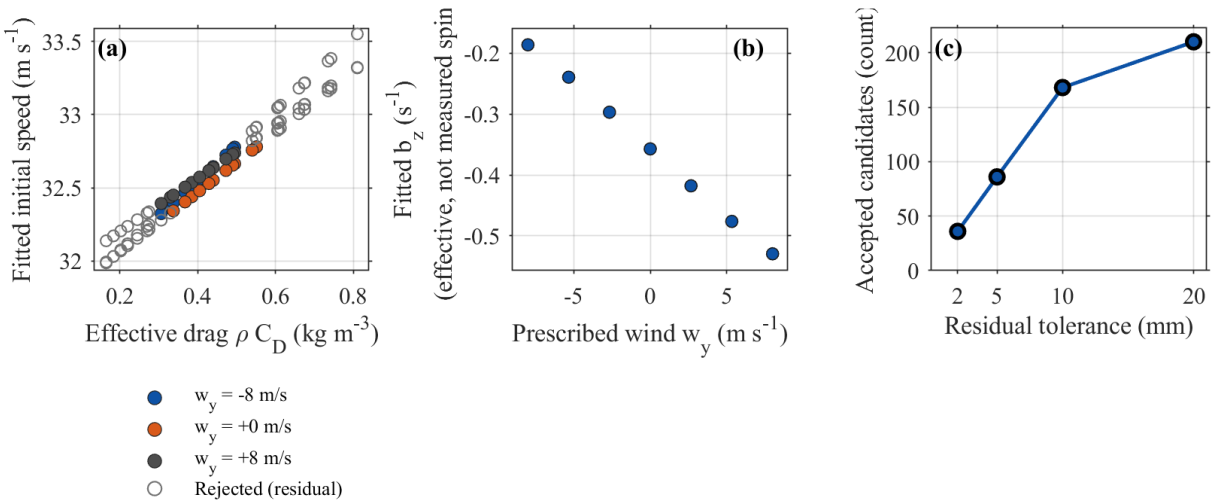


Figure 3: Parameter coupling and screening: density–drag product against fitted launch speed for three wind slices (left); wind against the fitted b_z for accepted trajectories (middle); accepted count against the allowed error (right). All counts refer to the listed parameter choices.

The goal-plane view in Figure 4 shows the range of crossing points from the accepted trajectories. Colour identifies the specified drag coefficient; the number of plotted points in a region carries no probability weight.

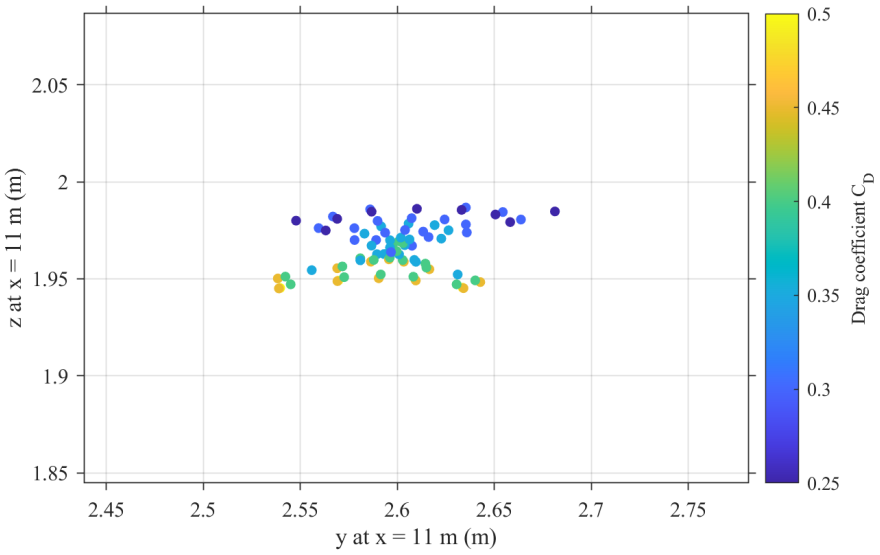


Figure 4: Goal-plane crossings of the accepted trajectories. Colour marks the listed drag coefficient. The spread is a property of the sampled parameter choices, not a probability distribution.

A separate interactive view allows wind in $[-15, 15]$ m s^{-1}, density in $[0.90, 1.40]$ kg m^{-3} and $C_D \in [0.08, 0.80]$; this wider exploration is distinct from the paper grid and uses the same screening.

5 Numerical checks and scope

This section lists the checks made inside MATLAB and the limits of the result.

All calculations use the same equations and event definitions. The three step sizes differ only in the maximum step of the integration, so they measure numerical sensitivity rather than a new

model. Using the balance derived in Eq. (11), we check the accumulated residual

$$R_E(t) = E(t) - E(0) - \int_0^t (\mathbf{F}_D + \mathbf{F}_M) \cdot \mathbf{v}\, d\tau. \tag{23}$$

Event checks confirm that ground contact, the forward crossing of $x = 11$ m and any time-out are located and recorded separately. The largest change in the goal-plane endpoint among the three step sizes is 6.66×10^{-15} m. The maximum magnitude of this work-energy residual is 1.99×10^{-13} J. Checks with deliberately excessive launch speed and side force were rejected as intended.

These checks test the code and the internal balance. No independent measured full flight was available, so they do not confirm that this particular ball, field and camera obey the model. The reported crossings and ranges apply only to the listed parameter choices and the stated allowed error. They carry no probability and no guarantee outside the searched values. A wider allowed error accepts more trajectories, but it does not turn the sampled spread into a set that must contain the true landing point.

6 Conclusion

This section answers the problem.

From the two early positions, a single trajectory cannot be recovered when the wind and aerodynamic coefficients are unknown. For each listed environment we can still fit launch conditions that reproduce both positions, and those accepted trajectories cross the goal plane at different points. For the current example at wind 8.0 m s^{-1} and density 1.225 kg m^{-3}, the centre crosses at $(2.6087, 1.9598)$ m after 0.4042 s. Across the listed choices, 86 of 210 fits pass the 5 mm test, with crossings spanning $y \in [2.5384, 2.6812]$ m and $z \in [1.9451, 1.9867]$ m; two accepted trajectories differ by 0.1477 m. These ranges describe the sampled grid only and are not probabilities. More observation times, measured wind and a measured drag coefficient would narrow the answer.

References

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Report on the use of AI

AI tools assisted with code generation, numerical checks and manuscript drafting. Human team members must verify all adopted content and take responsibility for it.

Team number and the complete AI records are still to be added; this is not a final submission. The contest rules require the exact input and complete output of the AI tools used; that record must be appended and reviewed before submission.